

Linearity enhancement of linear frequency-modulated DFB semiconductor lasers based on smoothing algorithms

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Frequency-modulated continuous-wave (FMCW) light detection and ranging (LiDAR) offers considerable potential for high-precision ranging and velocity measurement, yet its accuracy is critically limited by the linearity of the laser frequency modulation. To overcome the poor nonlinearity correction of conventional predistortion methods at sweep transition points, this paper introduces a smooth predistortion iterative algorithm. The algorithm is realized in a frequency-modulation nonlinear correction system built on a field-programmable gate array (FPGA). In contrast to traditional predistortion, the proposed technique enables more convergence iterations, substantially reducing modulation nonlinearity. Experimental results show that after 2700 iterations, laser frequency nonlinearity for up-ramp and down-ramp sweeps dropped from 3.3% and 1.3% to 0.022%, respectively. The corresponding residual nonlinearity, expressed as $1 - r^2$, reached 8.458×10^{-7} (up-ramp) and 8.4208×10^{-7} (down-ramp). Ranging accuracy reached a maximum error of 9 mm. After correction with the two algorithms, the same target was imaged, and image entropy was introduced as the metric for evaluating imaging quality. The target was a hollow diffuse reflection cardboard. The imaging results should present two planes, with an image entropy of approximately 1 bit theoretically. However, affected by noise and ranging accuracy, the actual value will be bigger than 1 bit. Image entropy was 1.4063 bits for the proposed algorithm versus 1.9585 bits for the conventional one, confirming the superior image quality of the proposed approach. © 2026 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

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1. INTRODUCTION

Frequency-modulated continuous-wave (FMCW) light detection and ranging (LiDAR) integrates optical heterodyne detection with linear frequency modulation, enabling breakthroughs in high-precision ranging, velocity measurement, and anti-interference capabilities. Consequently, it has become a core technology in fields such as autonomous driving and remote-sensing monitoring [1–3]. The linearity of the laser's frequency modulation directly dictates the performance limit of the system. However, nonlinearities inherent in practical frequency modulation cause spectral broadening of the signal, introduce ranging errors, and obscure Doppler shift information, thereby severely limiting the system's resolution and accuracy [4]. Therefore, correcting frequency modulation nonlinearity is crucial for enhancing the performance of FMCW LiDAR systems. Experimental results demonstrate that real-time compensation can improve FMCW LiDAR ranging

accuracy to the millimeter level while maintaining system stability in high-speed dynamic scenarios [5].

The correction of frequency modulation nonlinearity is pivotal for enhancing FMCW LiDAR performance. Primary techniques include optical phase-locked loops (OPLL), equal-frequency resampling, and current-iterative predistortion [6,7]. While the OPLL offers real-time compensation but suffers from high complexity and cost, and resampling is constrained by the Nyquist theorem and auxiliary path requirements [8,9] iterative predistortion stands out for its simplicity and hardware efficiency. However, conventional iterative methods often face challenges with convergence and limited correction efficacy. Significant efforts have been made to advance iterative predistortion. Zhang *et al.* pioneered an iterative learning algorithm, demonstrating substantial nonlinearity reduction in a distributed feedback (DFB) laser within just five iterations [10]. Subsequently, Cao *et al.* introduced a fast-convergence

algorithm using an inverse function, achieving high linearity with only six iterations [11,12]. More recently, Chen *et al.* proposed a piecewise linear interpolation-based method, which simplified computations and attained excellent correction within five iterations and under one minute of processing [13]. Building upon these foundations, this work proposes an FPGA-based smooth predistortion algorithm to address the critical issue of iteration non-convergence, particularly at sweep transition points.

This paper presents an FPGA-based smooth predistortion algorithm for high-precision, rapid-convergence nonlinear correction in frequency-modulated continuous-wave lasers. The system utilizes the FPGA for integrated laser driver control, data acquisition, and signal processing. By smoothing abrupt transitions in the reference frequency curve, the algorithm effectively resolves iteration non-convergence issues in drive voltage optimization. Experimental results show that using this smoothed reference enables a DFB laser to complete 2700 correction iterations in just 0.135 s, producing a high-linearity source with 20 kHz repetition rate and 8.6 GHz sweep bandwidth. The nonlinearity was reduced from 3.3% to 0.022% for up-ramp and from 1.3% to 0.022% for down-ramp sweeps, with corresponding residual nonlinearities (characterized by $1 - r^2$ of 8.458×10^{-7} and 8.4208×10^{-7}). The implemented ranging system achieved a maximum error of 9 mm at 5 m distance. Furthermore, in FMCW LiDAR imaging applications, the LiDAR system employing the smoothing correction algorithm achieved an image entropy of 1.4063 bits, whereas the traditional method yielded an image entropy of 1.9585 bits—indicating that the images acquired by the LiDAR system with the smoothing correction algorithm exhibit higher sharpness [14].

2. FUNDAMENTAL PRINCIPLES

A. Principle of FMCW Ranging

The operating principle of FMCW LiDAR relies on the path difference between the detection and local oscillator beams, as illustrated in the time-domain waveform below. The peak frequency of the beat signal spectrum maintains a deterministic relationship with the target distance [15]. By analyzing this beat signal, the instantaneous laser frequency can be derived, enabling accurate distance calculation.

Figure 1 illustrates the time-domain relationship between the local oscillation beam and the measurement beam, where T denotes the triangular-wave period, B represents the laser sweep bandwidth, and γ signifies the laser sweep rate. The sweep rate is expressed as

$$\gamma = \frac{2B}{T}. \quad (1)$$

Under ideal conditions, the beat frequency produced by mixing the local oscillator light and the measurement light should be equal to the difference between their respective frequencies:

$$f_b = f_l(t) - f_p(t) = \gamma\tau. \quad (2)$$

In the equation, τ represents the delay time corresponding to the optical path difference between the local oscillator light

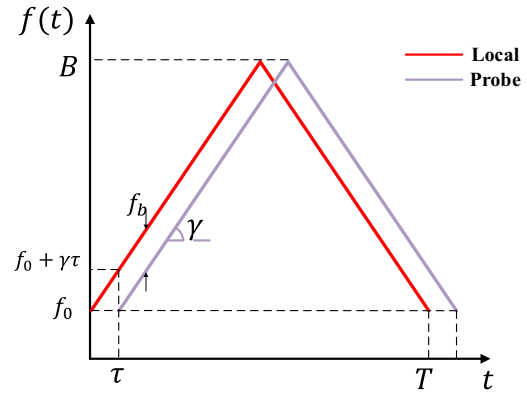


Fig. 1. Schematic diagram of the FMCW LiDAR ranging principle.

and the measurement light. Neglecting the fiber length between the circulator and the collimator, the optical path difference between the measurement light and the local oscillator light is twice the distance being measured. Hence, the measured distance is

$$D = \frac{f_b T c}{4B}, \quad (3)$$

where D denotes the measured distance and c represents the speed of light. The target distance is directly determined from the beat signal frequency. Through appropriate filtering and frequency domain analysis, the peak frequency of the beat signal can be accurately identified, enabling precise calculation of the target distance.

B. Principle of Iterative Predistortion

When a linearly ideal triangular drive voltage is applied to a DFB laser, the output optical frequency is affected by intrinsic factors, such as thermal hysteresis effects. These effects introduce nonlinear components into the output frequency, consequently degrading both ranging accuracy and imaging quality [16]. To analyze the linearity of the laser frequency sweep and implement iterative predistortion, the instantaneous laser frequency must first be acquired. While various techniques exist for measuring instantaneous laser frequency, including optical frequency combs, the predistortion iteration process only requires the relative frequency of the light source. Therefore, a fiber-based Mach–Zehnder interferometer (MZI) utilizing optical couplers is employed to measure the instantaneous laser frequency [17].

Figure 2 shows the block diagram of the iterative predistortion nonlinear correction system. The DFB laser driver circuit contains two distinct paths: one provides direct current (DC) bias to maintain constant laser emission, while the other applies a triangular modulation voltage generated by the FPGA. Within the FPGA, a linear ideal triangular waveform is synthesized, converted to an analog signal through a digital-to-analog converter (DAC), and used for laser frequency modulation. The output beam is directly coupled into a MZI with a relative delay τ between its two arms. The interferometer consists of two 50:50 couplers and a fixed-delay fiber segment. After interferometric mixing, the resulting beat signal passes through an adjustable optical attenuator to a balanced photodetector (BPD) for conversion to a voltage signal. This analog signal is

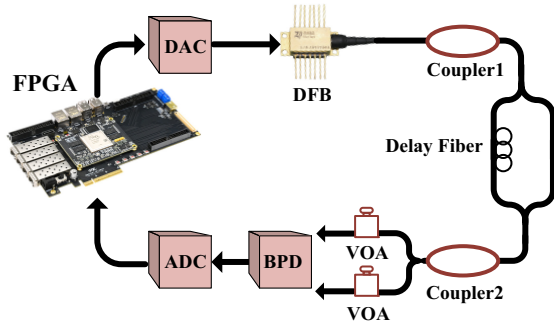


Fig. 2. Block diagram of the nonlinear correction system.

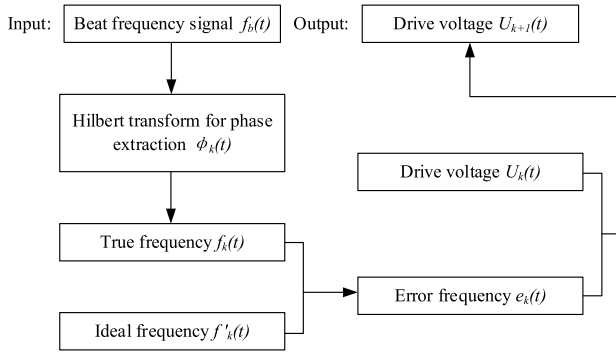


Fig. 3. Digital signal processing flowchart for the nonlinear correction experiment.

then digitized by an analog-to-digital converter (ADC) and acquired by the FPGA for processing. The instantaneous laser frequency is determined using the following relationship:

$$f(t) = f_c \pm \frac{1}{2\pi\tau n} \left[\arctan \left(\frac{f_b(t)}{HT(f_b(t))} \right) \right], \quad (4)$$

where $HT(f_b(t))$ denotes the Hilbert transform applied to the beat frequency signal, n represents the fiber refractive index, f_c denotes the laser carrier frequency, and τ signifies the delay time.

Figure 3 illustrates the detailed digital signal processing flowchart for the predistortion iterative algorithm. Taking the up-ramp sweep as an example, during the k^{th} iteration, the current drive voltage $U_k(t)$ is initially applied to the laser. Subsequently, the FPGA performs a Hilbert transform on the acquired beat signal $f_b(t)$ to extract its instantaneous phase, which is then utilized to calculate the actual instantaneous laser frequency $f_k(t)$. Following this, a linear fit is applied to the actual instantaneous frequency to derive the corresponding ideal reference frequency $f'_k(t)$. The relative frequency error for the current iteration is then determined by calculating the difference between the actual instantaneous frequency and the ideal frequency:

$$e_k(t) = f'_k(t) - f_k(t). \quad (5)$$

Converting the relative frequency error to relative voltage error, the driving voltage for the $k+1^{\text{th}}$ iteration is

$$U_{k+1}(t) = U_k(t) + p \cdot e_k(t), \quad (6)$$

where p denotes the error factor, the value of which influences the iteration speed and the effectiveness of the nonlinear correction.

C. Smooth Predistortion Iterative Algorithm

The nonlinear error of triangular-wave frequency sweep is predominantly concentrated within 20% of the sweep period adjacent to the sweep turning points, where the distortion acts as the dominant source of full-period sweep nonlinearity and directly determines the ultimate performance of the overall linearity correction [11,18]. Iterative correction performed exclusively within the 80% region of interest (ROI) cannot thoroughly eliminate the impact of turning-point distortion, as two types of global effects induced by such distortion propagate into the ROI interval. First, the abrupt voltage discontinuity at the turning points is the primary cause of transient thermal drift inside the laser chip. Since the time constant of the thermal effect is far longer than the sweep period, the resulting frequency drift is full-period and global, which inevitably leads to frequency deviations within the ROI. Second, the abrupt voltage transition at the sweep turning points excites carrier relaxation oscillations, whose decay tail extends into the leading and trailing segments of the ROI, leaving residual nonlinear errors originating from the turning-point distortion at the start and end of the ROI interval.

Iterative compensation targeting only the voltage points within the ROI can merely compensate for the residual error inside the ROI, but cannot eliminate the root cause of the distortion at the sweep turning points. The errors at the leading and trailing ends of the ROI will persist, resulting in slope discontinuities and abrupt waveform changes at the junction between the ROI and turning-point regions of the voltage waveform. Such abrupt changes in the driving waveform will induce new frequency distortion and high-frequency nonlinear noise in the subsequent sweep cycle, further degrading the correction performance within the ROI. Furthermore, performing iteration with only the error inside the ROI will cause forced abrupt truncation of the error signal at the ROI boundaries, generating discontinuous singularities. This will trigger oscillations in the iteration process, reduce the convergence rate, and even lead to iterative divergence, ultimately deteriorating the overall correction performance.

To break the vicious cycle of high-frequency noise injection caused by the mathematical sharp corners of the ideal triangular wave, the target frequency sweep curve must be modified to be physically realizable for the laser. The proposed smooth predistortion iterative algorithm achieves this by abandoning the continuous linear fitting approach. Instead, it determines the ideal frequency through direct calculation based on the sweep bandwidth and modulation period and strategically applies a smoothing operation at the sweep transition points.

Initially, the algorithm acquires the actual time-frequency curve using the same method as the traditional approach, deriving the instantaneous laser frequency through the Hilbert transform. Next, rather than performing a new linear fit for each iteration, the ideal instantaneous frequency for the main linear region (taking the up-ramp sweep as an example) is directly defined:

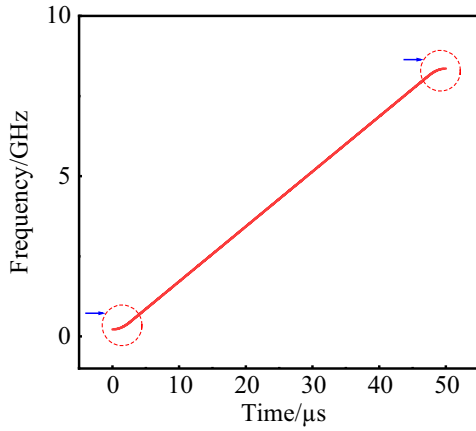


Fig. 4. Smoothed time-frequency curve of the ideal frequency.

$$f_{\text{ideal}}(t) = \frac{B}{T_s} t, \quad (7)$$

where B is the laser sweep bandwidth and T_s denotes half the period of the modulation triangular wave. To prevent the relative errors at both inflection point regions from exhibiting opposite polarities and large magnitudes, smoothing operations are applied to the ends of $f_{\text{ideal}}(t)$. This work utilizes a quadratic function for smoothing $f_{\text{ideal}}(t)$:

$$f_d(t) = \begin{cases} f_{\text{ideal}}(t), & n \leq t \leq T_s - n \\ f_{\text{ideal}}(t) + at^2 + bt, & t < n \cap t > T_s - n \end{cases}, \quad (8)$$

where n defines the smoothing region length, and a , b , and c are fitting coefficients.

Figure 4 shows the resulting smoothed ideal frequency. For the k^{th} iteration with drive voltage $U_k(t)$, the beat signal $f_b(t)$ acquired by the ADC is processed through the FPGA to obtain the instantaneous laser frequency $f_k(t)$. The relative error is computed as $e_k(t) = f_d(t) - f_k(t)$. The drive voltage for the next iteration is then updated as $U_{k+1}(t) = U_k(t) + p \cdot e_k(t)$, where p represents the error coefficient determined empirically from the experimental setup.

3. EXPERIMENTAL SETUP AND RESULTS

A. Non-Linear Correction Experiment

The experimental setup for the nonlinear correction is illustrated in Fig. 2. The core digital signal processing module is implemented on a XILINX ZYNQ XC7Z100 FPGA running at a clock frequency of 200 MHz. In this architecture, an ideal triangular wave is generated digitally by the FPGA and converted into an analog signal using a 16-bit DAC (AD9122) with a maximum sampling rate of 500 MSPS for frequency modulation of the DFB laser. The generated driving voltage has a period of 50 μs (corresponding to a repetition rate of 20 kHz) and a peak-to-peak amplitude of 1.75 V. Meanwhile, the optical beat signal detected by the balanced photodetector is digitized by a 14-bit ADC (AD9643) operating at 250 MSPS and then fed back to the FPGA for real-time processing. The measured spectrum of the swept laser source is shown in Fig. 5. With a sweep bandwidth of 8.6 GHz and a sweep period of 50 μs , substitution

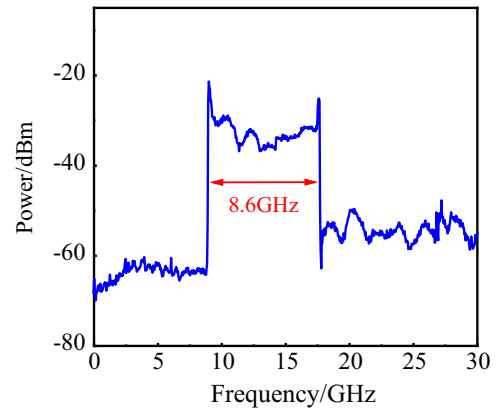


Fig. 5. Spectral diagram of the DFB laser modulation bandwidth.

of these parameters into Eq. (1) yields an ideal theoretical sweep rate of 3.44×10^5 GHz/s.

The experimental setup incorporated a MZI with a 4 m fiber delay line, providing a relative delay of 19.47 ns. Figure 6(a) displays both the uncorrected beat signal and its corresponding initial drive voltage. The Fourier transform of the pre-calibration beat signal, shown in Fig. 6(b), reveals a broad spectral peak with a full width at half-maximum (FWHM) of 2.1 MHz due to frequency sweep nonlinearities. The instantaneous frequencies and residual errors for the pre-calibration up-ramp and down-ramp sweeps are shown in Figs. 6(c) and 6(d), respectively, using an 80% ROI. Before correction, the nonlinearity values characterized by $1 - r^2$ were 1.7×10^{-2} for the up-ramp and 2.76×10^{-3} for the down-ramp, with corresponding residual errors of 330.5 and 130.5 MHz.

As can be seen from Figs. 6(c) and 6(d), the variation trends of the error curves for the up-ramp and down-ramp sweeps before correction are opposite. The error curve of the up-ramp sweep interval bends downward, with the error changing from positive to negative and then back to positive, while the error curve of the down-ramp sweep interval bends upward, with the error changing from negative to positive and then back to negative. At the end of the up-ramp sweep, the frequency error is positive, which requires a reduction in the driving voltage to minimize the error. Conversely, at the start of the down-ramp sweep, the frequency error is negative, which requires an increase in the driving voltage. After multiple iterations, this will lead to an abrupt change in the triangular-wave driving voltage at the sweep turning point, which impairs the iterative convergence and degrades the nonlinear correction performance.

Correction of the DFB laser was performed based on the conventional iterative predistortion algorithm. Figure 7(a) shows the beat signal and the corresponding driving voltage signal after 2700 iterations. As stated in the above theoretical analysis, the opposite error polarities of the up-ramp and down-ramp sweeps cause a jump in the sweep error at the sweep turning points, which further leads to an abnormal step-like discontinuity of the triangular-wave driving voltage at the turning points and impairs the final nonlinear correction performance. Figure 7(b) displays the Fourier transform spectrum of the corrected beat signal, showing an increase in peak intensity from -9.87 dBm before correction to 1.6 dBm after correction. The instantaneous frequencies and residual errors for the up-ramp and

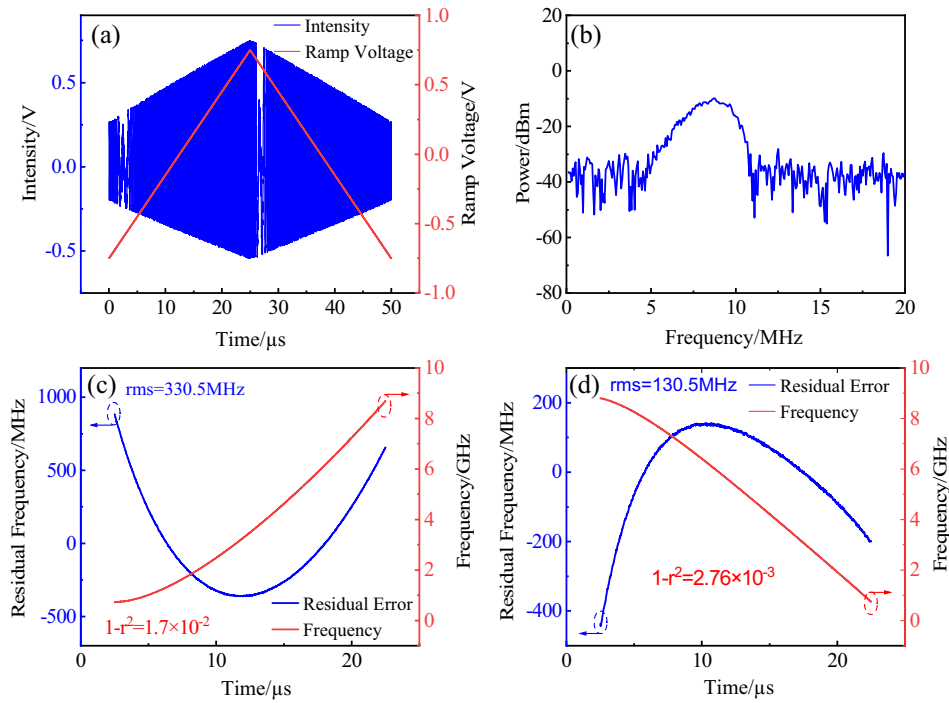


Fig. 6. (a) Initial drive voltage and beat frequency signal waveform before correction; (b) spectrum of beat frequency signal before correction; (c) residual error before correction for up-ramp; and (d) residual error before correction for down-ramp.

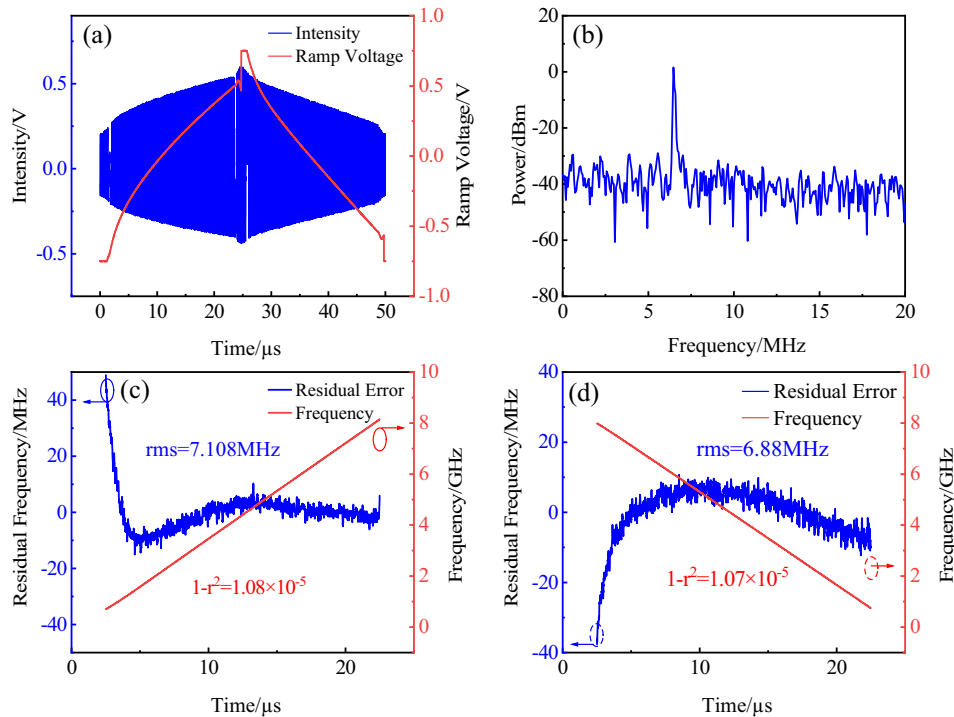


Fig. 7. Traditional linear fitting predistortion iterative algorithm. (a) Drive voltage and beat frequency signal waveform after correction; (b) spectrum of the beat frequency signal; (c) residual error after correction for up-ramp; and (d) residual error after correction for down-ramp.

down-ramp after linear fitting correction are shown in Figs. 7(c) and 7(d), respectively. After correction, the $1 - r^2$ values decreased to 1.08×10^{-5} and 1.07×10^{-5} , with corresponding residual errors of 7.108 and 6.880 MHz. The nonlinearities were reduced to 0.0826% and 0.08%, respectively.

Using the same nonlinear correction system, the DFB laser was corrected with the smooth predistortion iterative algorithm, reaching optimal linearity after 2700 iterations. Figure 8(a) shows the beat signal and corresponding drive voltage after smooth predistortion correction. The drive voltage waveform

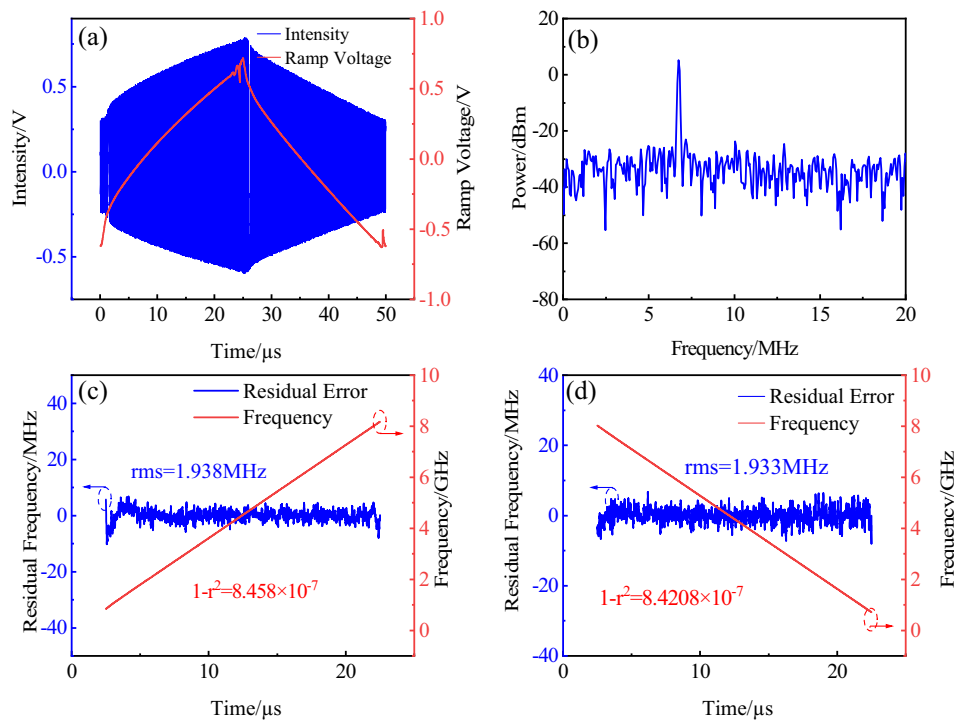


Fig. 8. Smooth predistortion iterative algorithm. (a) Drive voltage and beat frequency signal waveform after correction; (b) spectrum of the beat frequency signal; (c) residual error after correction for up-ramp; and (d) residual error after correction for down-ramp.

exhibits curvature with minor spikes at the peaks outside the ROI, which do not affect the convergence or final linearity. Figure 8(b) shows the Fourier spectrum of the corrected beat signal, demonstrating a FWHM below 0.12 MHz and a peak intensity improvement from -9.87 to 5.17 dBm. The instantaneous frequencies and residual errors after smooth nonlinear correction are shown in Figs. 8(c) and 8(d) for the up-ramp and down-ramp, respectively. The corrected $1 - r^2$ values decreased to 8.458×10^{-7} and 8.4208×10^{-7} , with residual errors of 1.938 and 1.933 MHz. The nonlinearity was uniformly reduced to 0.022% .

Figures 9(a) and 9(b) show the time-domain waveforms of the corrected beat signals obtained by the conventional iterative predistortion algorithm and the smooth iterative predistortion algorithm, respectively. As can be seen from Fig. 9(a), due to the step-like discontinuity of the triangular-wave driving voltage at the sweep turning points introduced by the conventional iterative predistortion algorithm, the beat signal after multiple iterations exhibits obvious high-frequency components at the turning points. Furthermore, the range affected by these high-frequency components expands continuously as the number of iterations increases. The presence of high-frequency components causes the instantaneous frequency extracted via the Hilbert transform to be higher than the true frequency of the swept light source, thus degrading the nonlinear correction performance. In contrast, as shown in Fig. 9(b), the high-frequency components in the beat signal corrected by the smooth iterative predistortion algorithm are significantly suppressed.

Figure 10 shows the experimental results of nonlinear correction for the same DFB laser using the two algorithms. Notably, after 3000 correction iterations, the nonlinearity

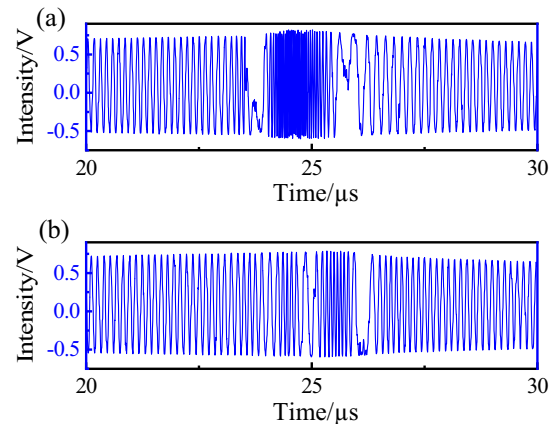


Fig. 9. Beat frequency signals in (a) traditional linear fitting predistortion iterative algorithms and (b) smooth predistortion iterative algorithms.

corrected by the conventional linear-fitting iterative predistortion algorithm begins to degrade. As the number of iterations increases, the nonlinearity of the laser frequency corrected by this linear-fitting method exhibits an overall trend of initial decrease followed by subsequent increase, and ultimately fails to achieve stable convergence. In contrast, when using the smooth iterative predistortion algorithm proposed in this paper, the nonlinearity of the corrected laser frequency shows distinct convergence characteristics, and this convergent state can be stably maintained even when the number of iterations is increased to 3600.

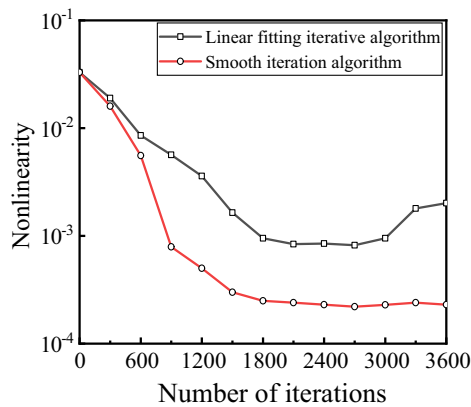


Fig. 10. Sweeping frequency nonlinearity of two algorithms as a function of the iteration number.

B. Distance-Measuring Experiment

To validate the effectiveness of the nonlinear correction, we conducted an FMCW-based ranging experiment. The system block diagram in Fig. 11(a) shows the FPGA driving a DAC to generate a 20 kHz triangular wave for laser frequency modulation.

The linearized laser output is amplified by an erbium-doped fiber amplifier (EDFA) and split by a 99:1 fiber coupler. The higher-power beam enters port 1 of a spatial circulator, is collimated at port 2, and illuminates a diffuse target. The return light from port 3 interferes with the 1% reference beam in a 50:50 coupler. The resulting beat signal is detected by a BPD, sampled by an ADC, and processed by the FPGA. The physical setup is shown in Fig. 11(b).

Ranging experiments were performed using the system in Fig. 11(a) with the same laser source corrected by both the smooth and linear fitting predistortion algorithms. Beat signals from targets at distances from 0.5 to 5 m (in 0.5 m increments) were acquired and processed by fast Fourier transform (FFT), producing the ranging spectra in Figs. 12(a) and 13(a). The measured distances were calculated from spectral peaks using

Eq. (3) after system delay subtraction, resulting in origin-passing calibration curves. Figures 12(b) and 13(b) show the corresponding ranging errors, with maximum values of 9 mm for the smooth algorithm and 16 mm for the linear fitting algorithm.

C. Imaging Experiment

To evaluate the imaging performance improvement enabled by the smooth predistortion algorithm, we constructed a coaxial FMCW LiDAR imaging system. Both imaging experiments utilized the same high-linearity DFB laser with 20 kHz triangular-wave repetition rate and 8.6 GHz sweep bandwidth, corrected by the two competing algorithms.

A LiDAR imaging system was implemented by adding a two-dimensional beam steering mirror after the collimator in the ranging system of Fig. 11(a). The imaging target (Fig. 14) is a diffusely reflective cardboard positioned 1.1 m from the steering mirror and 0.2 m from the background wall.

We imaged five letters at 128×128 pixel resolution. For each pixel, the detected and local oscillator lights were mixed, passed through the BPD and ADC, and processed by the FPGA for direct distance calculation. Figure 15(a) shows the result after smooth predistortion correction, demonstrating uniform 3D imaging density with minimal noise and well-defined edges. In contrast, Fig. 15(b) reveals significantly increased noise levels with the conventional linear fitting algorithm, though target edges remain discernible.

Image entropy was used to quantitatively evaluate imaging quality. Based on information theory, entropy measures the dispersion and uniformity of gray-level distributions. Higher entropy indicates more dispersed distributions and richer texture details, while lower entropy reflects more concentrated pixel distributions. For our two-plane imaging data, lower entropy corresponds to better image quality. Figures 16(a) and 16(c) show raw data images from the smooth and traditional algorithms, respectively, while Figs. 16(b) and 16(d) display their

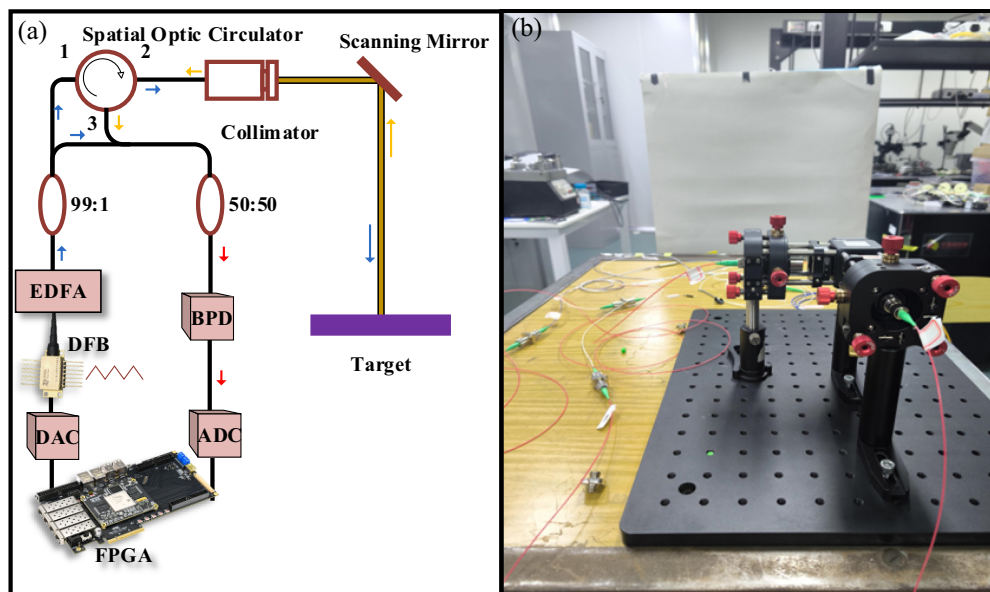


Fig. 11. (a) Block diagram of the ranging experiment system and (b) physical drawing of ranging experimental device.

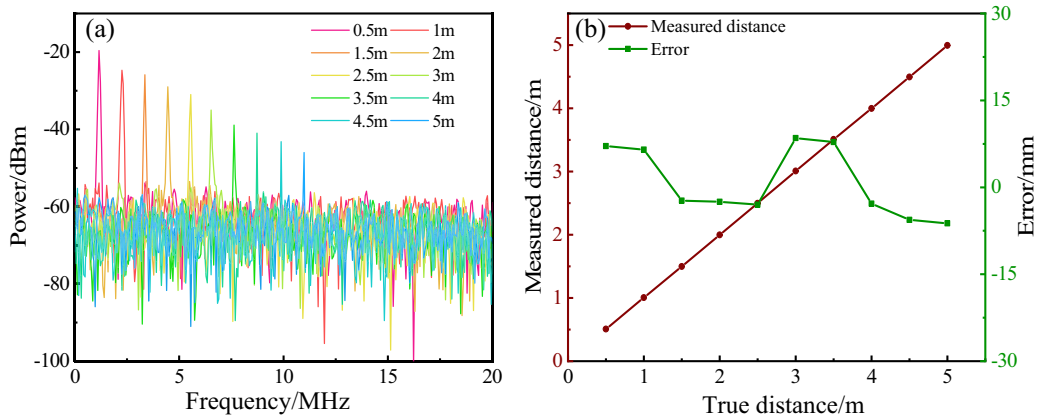


Fig. 12. Smooth predistortion iterative algorithm. (a) Correction ranging spectrum plots at different distances and (b) measurement distance versus true distance error plot.

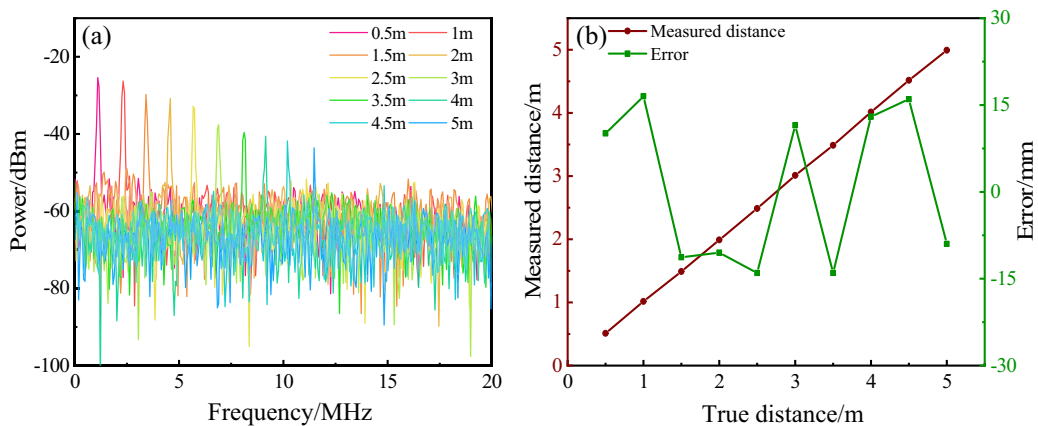


Fig. 13. Traditional linear fitting predistortion iterative algorithm. (a) Correction ranging spectrum plots at different distances and (b) measurement distance versus true distance error plot.

grayscale conversions after quantization and linear mapping. The calculated entropy values are 1.4063 bits for the smooth algorithm and 1.9585 bits for the traditional algorithm, confirming the superior performance of our approach. According to Ref. [14], lower entropy values denote a more concentrated distribution of image gray-level values, while higher entropy values correspond to a more dispersed distribution. The image information obtained from the experiment contains only two distance values (corresponding to two gray-level values in the image); theoretically, the image entropy should be approximately 1 bit under ideal conditions, meaning that the closer the entropy is to 1 bit, the better the image quality. However, affected by noise and ranging accuracy, the actual entropy value will increase to some extent. These imaging experiments validate the practical efficacy of the proposed algorithm in real LiDAR systems, supporting high-precision 3D imaging applications.

4. DISCUSSION

Recent advancements have demonstrated the significant potential of on-chip integrated asymmetric Mach–Zehnder interferometers (AMZIs) for real-time calibration in frequency-modulated continuous-wave (FMCW) LiDAR systems

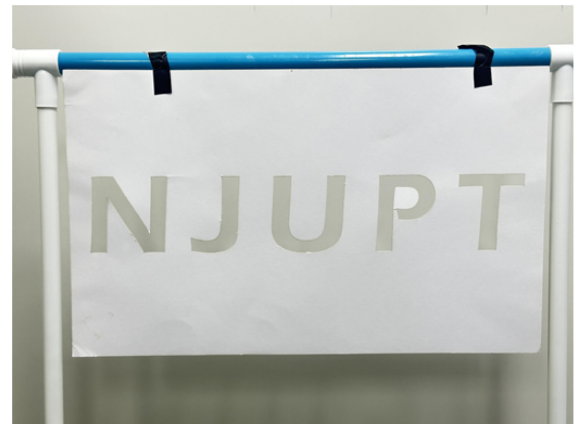


Fig. 14. Photograph of imaging target (diffuse reflection cardboard).

[19,20]. Previous studies have implemented predistortion-based iterative correction systems on compact chips via silicon photonic integration technology, while the fundamental predistortion principle remains consistent with that adopted in this work. The FPGA-based smooth iterative predistortion algorithm proposed in this paper exhibits excellent compatibility

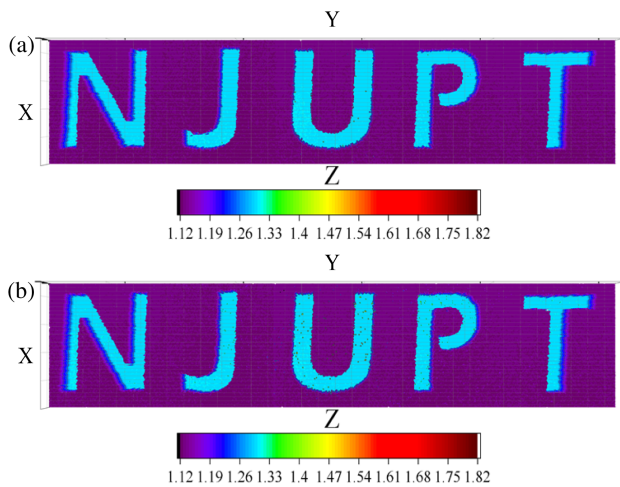


Fig. 15. (a) Image quality following correction by the smooth predistortion iterative algorithm and (b) image quality following correction by the traditional linear fitting predistortion iterative algorithm.

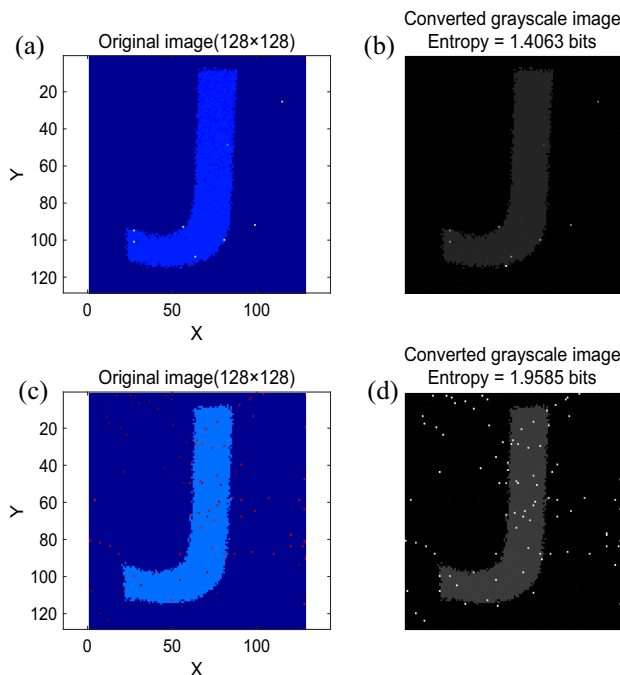


Fig. 16. (a) Raw data image produced by the smooth predistortion iterative algorithm; (b) grayscale conversion image produced by the smooth predistortion iterative algorithm; (c) raw data image produced by the traditional linear fitting predistortion iterative algorithm; and (d) grayscale conversion image produced by the traditional linear fitting predistortion iterative algorithm.

with such integrated photonic architectures. Since the algorithm processes the down-converted electrical beat signal in the digital domain, it can be directly used as the digital signal processing core of on-chip calibration engines without requiring any modification to the integrated photonic chip layout.

However, transitioning from a fiber-based MZI to an on-chip AMZI introduces specific limitations that must be addressed. Silicon photonic delay lines typically provide much shorter

optical delays (e.g., 8.8 ns in [19] and 7.56 ns in [20]) compared with the 19.47 ns relative delay offered by the 4 m fiber delay line used in our experimental setup. Fundamentally, a reduced delay time proportionally lowers both the beat signal frequency and accumulated phase, which inherently degrades the phase extraction resolution of the Hilbert transform process. To effectively compensate for this physical constraint imposed by on-chip integration, the system modulation parameters should be scaled accordingly—for example, by expanding the sweep bandwidth to restore the beat signal to the optimal frequency range. Furthermore, by simultaneously adjusting the error coefficient and smoothing interval length in the proposed algorithm to match the updated modulation parameters, the system can fully overcome the limitations induced by short delay lines, ensuring that the algorithm retains its fast convergence and high-linearity correction performance in highly integrated, chip-scale FMCW LiDAR applications.

5. CONCLUSION

This paper presents a smooth iterative predistortion method implemented on a FPGA platform. The method enables development of a linear light source operating at 20 kHz repetition frequency with 8.6 GHz sweep bandwidth. After correction, the frequency nonlinearity was reduced from 3.3% and 1.3% to 0.022% for up-ramp and down-ramp sweeps, respectively. A ranging system built around this linearized source achieved maximum errors of 9 mm within 5 m range. Unlike traditional linear fitting approaches, our FPGA-based solution integrates laser driver control, data acquisition, and real-time processing while implementing smooth correction of laser modulation signals for fast-converging adaptive linearization. The key innovation addresses iteration non-convergence through smoothing processing at inflection points of the reference frequency curve. Experimental results confirm that the smoothed reference waveform maintains drive voltage convergence and excellent linearity through 2700 iterations. Comparative ranging and imaging experiments demonstrate the significant advantage of our smooth predistortion algorithm over conventional methods in both accuracy and image quality.

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